Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Bachelor of Technology GitHub Profile
Shri Ramdeobaba College of Engineering and Management, Nagpur LinkedIn Profile

1 Education

[leftmargin=*,labelsep=0mm]

Bachelor of Technology in Computer Science and Engineering(Cyber Security) *2020-24*

Shri Ramdeobaba College of Engineering and Management, Nagpur CGPA: xx

2 Personal Projects

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Web Based Facial Authentication(Liveness Detection)

A website based facial authentication system, implemented using a Chrome Extension.

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- Facilitating users' logins to websites without having to remember their credentials
- Used Live detection techniques to create high order security.
- Technology Used: Python, Reactjs, Bootstrap.

Realtime Chat App

A react based web application which allow users to chat in real time.

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- Used Firebase Authentication(SDK) to facilitate authentication & Cloud Firestore to store data.
- Technology Used: Reactjs, Firebase, Bootstrap, HTML.

Covid-19 Tracker

Daily and weekly updated statistics tracking the number of COVID-19 cases, recovered, and deaths.

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- Tracking world-wide cases using google maps and live API stats and datasets.
- Technology Used : JavaScript , CSS, HTML, API.

3 Experience

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AWS Cloud Virtual Internship *May - July 2023*

AICTE-Eduskills Online

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- In-depth understanding of AWS cloud computing services, including EC2, S3, RDS, Lambda, IAM, VPC, and more.

- Proficient in designing, deploying, and managing fault-tolerant, highly available, and scalable AWS solutions.
- Strong knowledge of architectural best practices, such as AWS Well-Architected Framework, security, performance, and cost optimization.
- Hands-on experience in cloud infrastructure provisioning, monitoring, and automation using AWS Management Console and AWS CLI.

- **Palo Alto Cybersecurity Virtual Internship** *Dec 2022 - Feb 2023*
AICTE-Eduskills Online
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  - Learned the fundamentals of Security Operations Center (SOC).
  - Learned basics of Network & Cloud Security.

4 Technical Skills and Interests

[leftmargin=0.05in, label=] **Languages**: C/C++, Python, Javascript, HTML+CSS **Libraries** : C++ STL, Python Libraries, ReactJs**Web Dev Tools**: Nodejs, VScode, Git, Github **Frameworks**: ReactJs **Cloud/Databases**:MongoDb, Firebase, Relational Database(mySql) **Relevent Coursework**: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Software Engineering. **Areas of Interest**: Web Design and Development, Cloud Security.**Soft Skills**: Problem Solving, Self-learning, Presentation, Adaptability

5 Positions of Responsibility

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On Desk Registrations Volunteer Aarhant Cyber Week Event - RCOEM, Nagpur *Oct - Dec 2022*
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- Helped to attract close to 300 attendees to the event.
- Collected over Rs. 20,000 in entry fees for different activities.